1 Phase 1: Project Setup & Intelligent Discovery

1.1 1. Dashboard & Global Inventory

Smart Overview: Upon logging in, users create a project (e.g., "Feature Selection in Cyber-Physical Systems"). The dashboard displays a real-time status overview:

- Total papers read / deeply analyzed.
- Pending / Unread papers queue.
- Total active methodology clusters.
- Top Keywords (rendered dynamically as a Filterable Chip List or Interactive Word Cloud).

Recommendations:

1. **Thesis Readiness Velocity Score:** Integrate an animated circular progress ring tracking overall thesis extraction completeness (e.g., "Benchmark Matrix 84% Complete: 42/50 papers deeply extracted, 14 critical research gaps mapped").
2. **Pillar Quick-Switch Tabs with Dynamic Badges:** Include top-level filter tabs corresponding to methodology pillars (e.g., Pillar 00: Surveys, Pillar 01: Filters, Pillar 02: Wrappers, Pillar 03: Embedded, Pillar 04: Hybrids, Pillar 05: Specialized) with live paper count badges.
3. **Living Review Alert Feed (Elicit & ResearchRabbit Engine):** Add a persistent notification bell delivering real-time preprint/paper alerts when new studies matching the project's keywords appear on arXiv, IEEE, or PubMed with a 1-click "Add to Screening Queue" action.
4. **Global Semantic Paper Search:** Enable natural language queries against OpenAlex (250M+ papers) and CrossRef to discover seminal studies without requiring existing local PDFs.

1.2 2. Dynamic Clustering

Overview: Users can create custom clusters/categories tailored to their thesis topology. Complete flexibility: Clusters can be:

- Added,
- Renamed,
- Deleted at any time.

Recommendations:

1. **Color & Icon Badge Customizer:** Allow users to assign distinct hex badge colors (e.g., #38bdf8 for Filters, #a855f7 for Wrappers, #10b981 for Embedded, #f59e0b for Hybrids, #f43f5e for Specialized) and academic domain icons ([Energy], [Security], [Healthcare], [Genomics], [Network], [Vision], [Analytics], [Notes]).

2. **Archetype Mathematical Formulation Attachment:** Enable pinning a foundational formula and variable dictionary to the cluster header so all papers in that category automatically inherit the theoretical baseline:

$$\gamma_B(D) = \frac{|\mathrm{POS}_B(D)|}{|U|} \tag{1}$$

where $\mathrm{POS}_B(D)$ denotes the positive region of decision attribute D with respect to attribute subset B, and $|U|$ is the universe cardinality.

3. **2-Level Nested Sub-Taxonomy Trees:** Support sub-categories within major clusters (e.g., Wrapper Methods → Swarm Intelligence [PSO/GWO] vs. Evolutionary Algorithms [GA/DE]).

4. **Safe Cluster Archiving & Paper Re-routing:** When deleting a cluster, prompt a modal offering to reassign papers to another cluster or move them to an "Unassigned Archive" without deleting underlying PDF data.

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2 Phase 2: PRISMA-Compliant Data Collection & AI Screening

2.1 3. BYOData (Bring Your Own Data) & DOI Auto-Fetching

Overview:

- **Bulk Upload:** Users can drag-and-drop or browse their directories to upload multiple PDF files simultaneously.
- **DOI Integration:** If a user inputs a DOI or URL, the system fetches metadata (Title, Abstract, Year, Citation Count) via Crossref and Semantic Scholar APIs.
- **Pairing Guarantee:** The system does not download the PDF automatically. The user browses and uploads the PDF file, and the system pairs it with the fetched DOI metadata.
- **Agentic AI API Capabilities:**
  - Auto-filling metadata of the columns.
  - Setting its cluster for automatic routing.
  - Adding normalized domain keywords.

Recommendations:

1. **Intelligent Deduplication & Auto-Merge Engine (Rayyan-style):** Run automated fuzzy string matching (Levenshtein distance on normalized titles + exact DOI collision detection) during bulk upload. Display a "Duplicate Reconciliation Modal" allowing 1-click merging of duplicate records from multiple repository imports.

2. **Agentic AI Auto-Classifier & Tagging Pipeline:** When a PDF is dropped:
   - An autonomous micro-agent computes cosine similarity between paper abstracts and active clusters, recommending the best cluster with a confidence score (e.g., "96% Confidence Match → Pillar 01: Filters").

- Auto-generates 5 normalized domain keywords (e.g., #information-gain, #mrmr, #high-dimensional).
- Resolves publication venues (IEEE TPAMI, Elsevier KBS, Springer, Nature, ACM).

3. **Local PDF Text & Metadata Extractor:** Extract raw text, author affiliations, publication year, and structured abstract sections locally via `pdfjs-dist` to enable offline-first productivity.

2.2 4. AI PRISMA Screening (Inclusion / Exclusion Engine)

Overview:

- **Natural Language Criteria:** Before papers enter the "Unassigned" queue, users define filtering rules (e.g., "Exclude papers published before 2020, exclude survey papers, must evaluate on benchmark datasets").
- **Auditable Traceability:** The AI scans hundreds of papers in seconds, routing non-compliant papers to an Excluded List. To strictly follow PRISMA 2020 guidelines, it displays an exact quote or a generated Reason next to the excluded paper explaining why it was dropped.
- **Cluster Assignment:** Approved papers can be moved to specific clusters via:
  - Drag & Drop,
  - Dropdown selection,
  - A "Select to Transfer" batch menu.

Recommendations:

1. **5-Star AI Priority Relevance Ranking (Rayyan-style):** After the user manually screens the first 5–10 papers, an active learning classifier predicts inclusion probability for remaining papers, tagging them with 1 to 5 star relevance badges so high-impact papers float to the top of the screening queue.
2. **Interactive PRISMA 2020 Flowchart Generator:** Dynamically render a publication-standard PRISMA 2020 flow diagram (Identification → Screening → Eligibility → Included) with real-time numeric counters, exportable in 1-click as vector SVG, PDF, or Overleaf LaTeX TikZ figure.
3. **Color-Coded Inclusion/Exclusion Keyword Radar:** Auto-highlight user-defined positive keywords (e.g., benchmark, complexity, accuracy) in soft green and disqualifying terms (e.g., review only, qualitative, simulation only) in soft red within abstract screening cards.
4. **Student-Advisor Double-Blind Mode:** Support a blind screening toggle where co-authors/supervisors evaluate studies independently, revealing discrepancies and computing Inter-Rater Reliability (Cohen's Kappa κ) during final conflict resolution:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{2}$$

   where p_o is observed agreement and p_e is chance agreement.

3 Phase 3: The Smart Split-Screen Workspace (Deep Extraction)

3.1 5. Split-Screen Reading & Visual Extraction

Overview:

- Clicking a paper inside a cluster opens the Workspace.
- **Left Panel:** PDF Viewer (supports zoom, pan, and highlight).
- **Right Panel:** Dynamic Data Entry Panel (displays all columns, keyword tagging, and edit options).
- **Quick Actions:** Change cluster assignment or modify keywords directly from this view.
- **Visual Bounding Box:** Users can draw a box (crop) over diagrams, experimental tables, or math equations in the PDF. The selection is instantly saved as an image or extracted text in the corresponding cell on the right panel.

Recommendations:

1. **Math Equation Snipping & KaTeX OCR (SciSpace-style):** Drawing a bounding box over any mathematical formulation converts pixel data into valid LaTeX code rendered live via KaTeX, while automatically generating an editable symbol dictionary table ($symbol \to definition$).
2. **Benchmark Table-to-CSV Matrix Extractor:** Cropping experimental result tables (e.g., Accuracy %, F1-Score, Execution Time) parses table cells into structured JSON/CSV data ready for direct injection into the benchmark comparison matrix.
3. **Contextual Text-Selection Tooltip (Quick-Send Actions):** Highlighting any paragraph in the PDF opens a non-intrusive floating menu: `[Send to Core Intuition]`, `[Add to Key Strengths (Pros)]`, `[Add to Critical Gaps]`, `[Send to Math Formula]`.
4. **In-Viewer AI Thesis Copilot (SciSpace-style):** An inline explain button ("Explain for Thesis") that translates complex technical jargon into concise, academic language.

3.2 6. Dynamic Column Management & Split Logic

Overview:

- **Add & Split:** Users can create custom columns (which automatically sync across all papers in that cluster). Crucially, they can Split a column into sub-columns:
  - Splitting Complexity into Time Complexity (TC) and Space Complexity (SC).

Recommendations:

1. **Type-Aware Column Schemas:** Support diverse column data types:

- Text / Rich Text (Intuition, Limitations)
- KaTeX Math Formula (Formula math blocks)
- Multi-Tag Chips (Datasets, Classifiers, Domains)
- Numeric Metric (Accuracy %, Execution Time in seconds)
- Bullet Lists (Pros, Gaps, Open Opportunities)

2. **Cluster-Wide Schema Sync with Retroactive Migration:** Splitting or adding a column in one paper instantly propagates the schema to all papers in that cluster, with the AI retroactively splitting existing composite text into the newly formed sub-columns.
3. **Column Reordering & Visibility Toggles:** Allow drag-and-drop column reordering and checkbox-based visibility toggling to customize workspace density.

3.3 7. AI Auto-Fill & Human-in-the-Loop Verification

Overview:

- **Autonomous Ingestion:** As soon as a user adds a new column, the Research Agent reads the entire PDF and auto-fills extracted values.
- **Verification Badges:**
  - AI-populated data receives an [AI Generated] badge.
  - Once the user reviews the data and clicks the checkmark or edits the text, it converts to a permanent [Human Verified] badge.

Recommendations:

1. **Confidence Scores & Low-Confidence Flags:** Pair the [AI Generated] badge with an extraction confidence indicator (e.g., 96% High Confidence vs. 58% Low - Needs Review). Low-confidence items trigger a direct verification prompt.
2. **Custom Natural Language Prompt Builder (Elicit-style):** Allow users to customize the exact extraction prompt per column (e.g., "Extract the exact objective function minimized, including penalty terms" or "List all benchmark datasets with sample sizes").
3. **Human-in-the-Loop Dual Review Workflow:** Provide a "Verify All" batch auditor where researchers can cycle through AI-extracted snippets side-by-side with original PDF paragraphs in seconds.

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4 Phase 4: Master Matrix & High-Scale Analysis

4.1 8. Cluster-Based Dynamic Spreadsheet

Overview:

- Navigating back to the cluster's main page reveals a massive, dynamic Excel-like table capable of loading thousands of data points smoothly.
- **Inline Edit:** Users can click on any cell in the table to quickly update or edit values without needing to open the split-screen view.

Recommendations:

1. **Multi-Mode Presentation View Switcher:** Provide 4 switchable perspectives:
   - **Cards View:** Deep analytical cards with collapsible KaTeX variable drawers.
   - **Benchmark Table View:** High-density virtualized spreadsheet with multi-column sorting and pinning.
   - **Heatmap Matrix View:** 2D performance grid (Algorithms vs. Datasets vs. Accuracy).
   - **2D Litmaps Graph View:** Interactive citation timeline landscape.
2. **Universal Multi-Field Search & Filter Pills:** Real-time search across titles, authors, algorithms, mathematical symbols, dataset chips, and gap keywords.
3. **Column Aggregation & Statistics:** Automatic footer calculations showing total datasets evaluated, average accuracy, and metric distributions across the cluster.

4.2 9. Sentence-Level Traceability (Source Citation)

Overview:

- **Source Highlighting:** Clicking any value inside the Master Table or the right-side panel triggers the left PDF viewer to automatically scroll to the exact source. The specific sentence the AI extracted the data from is highlighted in yellow.

Recommendations:

1. **Pixel-Accurate PDF Anchor Coordinates:** Store page numbers and normalized bounding box coordinates $[pageNumber, x, y, width, height]$ alongside extracted data for sub-second jump-to-page navigation and precision yellow highlighting.
2. **Audit Verification Stamp & Quote Drawer:** Clicking an extraction badge opens a provenance drawer displaying: "Extracted from Page 6, Paragraph 3, IEEE TPAMI 2024. Verified by Mostafa Kamal on Aug 2026."
3. **Zero-Hallucination Academic Trust:** If an extracted claim lacks an exact verifiable sentence anchor in the PDF, flag it with an Unanchored Warning requiring researcher confirmation.

5 Phase 5: Insights, Report Generation & Sharing

5.1 10. Automated Unique Value Summary (Cluster Insights)

Overview:

- At the bottom of the table (or in a sidebar), the system analyzes column data and auto-renders a bulleted summary of unique values (e.g., "Datasets Used: NSL-KDD, MNIST" or "Key Limitations: High memory usage, Slow convergence").

Recommendations:

1. **Frequency Distribution Badges:** Display frequency count tags next to every unique element (e.g., NSL-KDD [28 papers], MNIST [21 papers], CIC-IDS2017 [12 papers]).
2. **Automated Research Gap & White-Space Visualizer:** Highlight unexplored algorithmic intersections (e.g., "Metaheuristic Optimization has been tested across 24 Healthcare studies, but 0 papers exist in Cybersecurity IoT — Prime Thesis Opportunity!").
3. **Consolidated Methodological Trade-Off Matrix:** Automatically cross-tabulate Pros vs. Gaps across all papers in the cluster to identify structural bottlenecks.

5.2 11. AI Research Reports (Automated Thesis Drafts)

Overview:

- **Generate Report:** Clicking this button prompts the AI to synthesize all papers in the cluster, custom column splits, and unique summaries to write a structured, 10–15 page Literature Review Draft.
- Every claim in the report includes an inline citation. Clicking the citation jumps directly to the specific line in the original PDF.

Recommendations:

1. **Modular Thesis Chapter 2 Outlines:** Offer pre-formatted academic chapter structures:
   - Section 2.1: Mathematical Archetypes & Theoretical Formulations
   - Section 2.2: Comprehensive Benchmark Taxonomy (Filter vs. Wrapper vs. Hybrid)
   - Section 2.3: Empirical Performance & Complexity Comparison Matrix
   - Section 2.4: Identified Limitations & Novel Thesis Contribution Roadmap
2. **Cross-Paper Multi-Document Synthesis Q&A (Elicit/SciSpace Engine):** An interactive research chat allowing multi-paper comparative queries (e.g., "Compare the computational time complexity bottlenecks of Swarm Intelligence vs. Rough Set methods on high-dimensional genomic data").

3. **Clickable Interactive In-Text Citations:** Every paragraph claim includes linked citations ([1], [2], Peng et al., 2005) that jump directly to the verified source quote in the split-screen PDF.

5.3 12. Custom Export & Transparent Sharing

Overview:

- **Selective Export:** Users can export data by cluster or select specific columns to download as Excel (.xlsx), CSV, RIS, or DOCX files.
- **Public Link:** Users can generate a Read-only or Editable link to share the entire split-screen dashboard with supervisors or co-authors — no account creation required for the guest.

Recommendations:

1. **Overleaf-Ready LaTeX Booktabs & BibTeX Exporter:** Export publication-ready LaTeX table snippets (`\toprule`, `\midrule`, `\bottomrule`) and a clean, validated .bib citation library ready for instant Overleaf integration.
2. **High-Resolution Vector SVG / 300 DPI PNG Graph Export:** 1-click download of the 2D citation network map and PRISMA 2020 flowchart formatted with legends for direct inclusion in thesis figures.
3. **Full Portable JSON Workspace Backup:** 1-click export/import of the entire project state, column schemas, extracted formulas, and annotations for backup or migration.

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6 Macro-Topology Engine (Litmaps + ResearchRabbit Integration)

6.1 13. Interactive 2D Citation Network & Algorithmic Lineage Trees

Overview:

- **2D Citation Landscape Canvas:** Interactive timeline visualization where X-axis = Publication Year (1991 → 2026), Y-axis = Citation Impact, and nodes are color-coded by Methodology Pillar.
- **Seed-Paper Snowballing Engine:** Designate a foundational seed paper (e.g., Peng et al. 2005 - mRMR) to automatically discover top backward references and forward citing papers via OpenAlex / Semantic Scholar.
- **Tri-Directional Literature Discovery Hops:**
  - Earlier Work (Backward): Foundational ancestral algorithms and theoretical baselines.

  - Similar Work (Lateral): Peer benchmark approaches on comparable datasets.
  - Later Work (Forward): Recent 2024–2026 algorithmic extensions and hybridizations.
- **Co-Authorship & Research Lab Collaboration Graph:** Visual network mapping top publishing authors and research groups in the field.

7 Business Model & Pricing Strategy

7.1 Subscription Tiers

Tier	Price	Target Capabilities & Inclusions	Deployment Mode
Basic	Free	Unlimited PDF uploads, manual data entry, KaTeX formula editor, standard CSV/BibTeX export.	Personal Workspace
Plus	$15 / month	AI PRISMA Screening, Dynamic Column Splitting, DOI Auto-Fetch, 500 AI Auto-Fill Credits/mo.	Managed Cloud
Pro	$40 / month	Sentence-Level Traceability, Visual Math OCR Bounding Box, AI Thesis Drafts (10+ pages), 2D Litmaps.	High-Performance Cluster
Enterprise	Custom	Unlimited team seats, university SSO, dedicated private LLM instances, dedicated API access, SLA support.	On-Prem / Dedicated VPC

Pricing Recommendations:

1. **Academic Student Annual Plan (50% Off):** Special verified .edu tier at $8/month (billed annually at $96/year) to capture global Master's and PhD student markets during thesis writing seasons.
2. **Pay-As-You-Go AI Token Top-Ups:** Small on-demand credit packs (e.g., $5 for 100 deep-extracted papers) for researchers working on massive scoping reviews without requiring enterprise contracts.
3. **Departmental / Lab Seat Packs:** Discounted multi-seat packages (5 seats for $150/mo) tailored for academic research labs and university departments.

8 System Compliance & Architectural Verification Matrix

— **End of Specification v5.0** —
Engineered for High-Precision Academic Synthesis & Thesis Acceleration

Workflow Dimension	**Compliance Standard**	**Operational Verification Target**
Systematic Review Audit	PRISMA 2020	Mandatory exclusion rationales with verbatim textual quotes.
Data Integrity	Zero-Hallucination	Sub-millisecond bounding box citation anchoring ($[p, x, y, w, h]$).
Mathematical Rigor	AMS / KaTeX OCR	Native LaTeX mathematical derivation and symbol variable dictionaries.
Taxonomy Scalability	Dual-Level Trees	Dynamic cluster lifecycle with safe migration and retroactive schema sync.
Interoperability	LaTeX / BibTeX / Word	Direct compilation with Overleaf booktabs and RIS export.